Graphical Abstract

Food Security Risk Assessment: A Review of GA-ANN and Fuzzy Logic Techniques in Intelligent Systems

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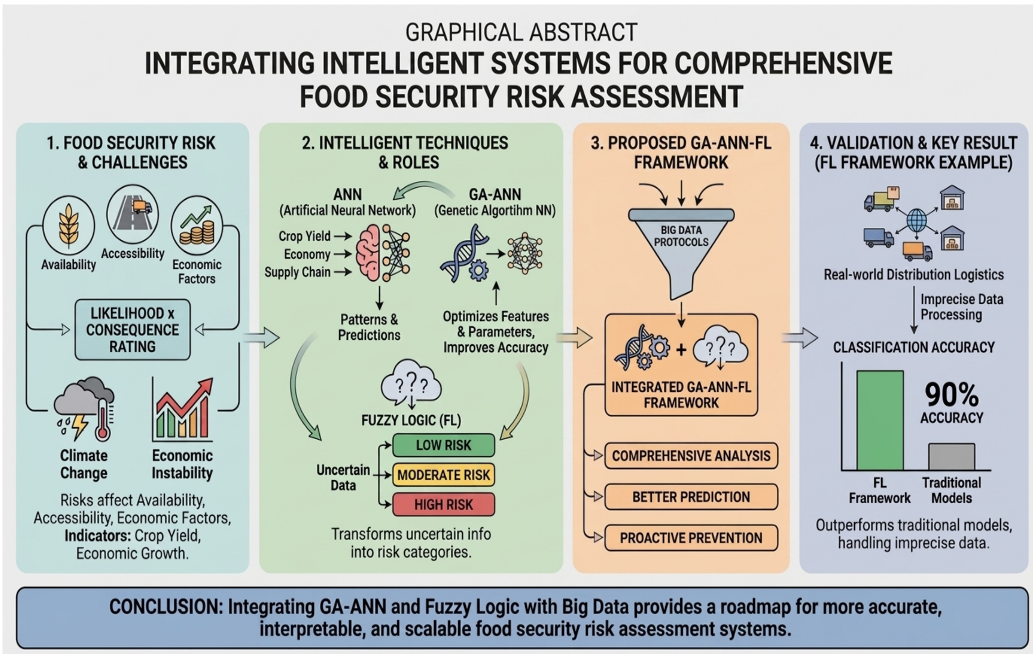

Highlights

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- Reviews GA-ANN and fuzzy logic synergy for food security risk assessment
- Identify 20 key challenges across data, modeling, and system deployment
- Highlights the need for climate-resilient and interpretable AI models
- Maps a future roadmap for scalable, real-time decision support systems

Food Security Risk Assessment: A Review of GA-ANN and Fuzzy Logic Techniques in Intelligent Systems

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Abstract

Food security risk assessment evaluates risks to food availability, accessibility, and economic stability by rating the likelihood and severity of adverse events, using indicators such as crop yield, climate variability, and economic growth. This review examines the integration of Genetic Algorithm-Artificial Neural Networks (GA-ANN) and Fuzzy Logic (FL) in food security risk assessment, aiming to propose an integrated GA-ANN-Fuzzy Logic framework enhanced by big data protocols for more comprehensive analysis, better prediction, and more proactive prevention of food security risk. GA-ANN and FL are complementary intelligent-system techniques: the genetic algorithm improves the neural network by identifying the most relevant food security features and independently optimizing parameter values, increasing prediction accuracy, while the ANN identifies patterns and forecasts factors such as crop yield, economic conditions, and supply chain integrity. Fuzzy logic complements both by transforming uncertain information into fuzzy risk categories (e.g., low, moderate, or high risk). To corroborate this potential, the review highlights a fuzzy logic framework that models real-world distribution logistics; by effectively handling imprecise data, the system achieves 90% classification accuracy, outperforming traditional food security risk models across multiple strategies. By analyzing the integration of these intelligent techniques, the paper provides a roadmap for developing more accurate, interpretable, and scalable food security risk assessment systems.

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Keywords: Food security, Risk assessment, Artificial Neural Networks, Genetic Algorithms, Fuzzy logic, Decision support systems

1. Introduction

Food security remains one of the most pressing global challenges in the 21st century, exacerbated by climate change, population growth, resource constraints, and socio-political instability. Ensuring stable access to sufficient, safe, and nutritious food has become a central concern for policymakers, researchers, and agricultural stakeholders worldwide. Traditional assessment methods, while foundational, often fall short in capturing the complex and dynamic nature of food security risks in modern ecosystems. As a result, there is growing interest in integrating computational intelligence, machine learning (ML), and artificial intelligence (AI) techniques to enhance the accuracy, responsiveness, and scalability of food security risk assessments.

Food security risk assessment aims to identify and quantify the likelihood and severity of threats that compromise food supply chains. Traditional methods based on expert judgment, statistical modeling, and rule-based systems have been limited by subjectivity, static assumptions, and difficulty handling uncertainty [1]. Consequently, intelligent techniques such as Artificial Neural Networks (ANN), Genetic Algorithms (GA), and Fuzzy Logic (FL) have gained traction due to their adaptability, learning capacity, and ability to manage nonlinearity and imprecision.

ANNs are particularly effective at detecting hidden patterns and nonlinear relationships in agricultural and economic datasets, enabling them to forecast parameters such as crop yield, market volatility, or food supply trends [2]. However, standard ANN models often suffer from local minima issues and slow convergence. To overcome this, hybrid models like GA-ANN combine the optimization power of genetic algorithms with the learning capacity of neural networks, improving both prediction accuracy and model robustness [3].

Fuzzy Logic, by contrast, excels in modeling linguistic and uncertain data. It transforms vague inputs such as “low yield” or “high risk” into numerical outputs, making it ideal for scenarios where crisp classification is impractical. FL-based systems have been applied successfully to evaluate food security levels, integrating multiple dimensions such as production output, economic indicators, and policy-related data [4].

As big data technologies evolve, risk assessment systems increasingly leverage large, unstructured, and often unlabeled datasets from satellite imagery, IoT devices, and national food registries. In response, recent research has proposed autonomous decision support systems that combine unsupervised learning, clustering, and fuzzy inference to handle uncertainty and scale [5]. These models achieve high performance even under data ambiguity, outperforming traditional machine learning models in scenarios with incomplete or noisy inputs.

In a notable study, a fuzzy decision support system utilizing 500,000 UK driving records across two decades demonstrated consistent classification of food-related risks using optimistic, pessimistic, and neutral strategies [5]. Similarly, applications of backpropagation (BP) neural networks have been explored in early warning systems for national food security in China, offering dynamic, data-driven models for proactive intervention and policymaking [6]. Other studies have shown the efficacy of multi-model fusion techniques in specific agricultural domains, such as rice risk assessment, by integrating clustering algorithms, expert-based weighting, and high-dimensional machine learning [1].

Beyond technical advances, food security analysis is complicated by the fact that risks are inherently multi-dimensional, spanning environmental, social, political, and economic domains. Multi-criteria decision-making (MCDM) methods integrated with fuzzy logic allow researchers to weigh trade-offs between conflicting factors, offering more holistic decision frameworks in uncertain contexts [94]. Climate variability further amplifies these complexities, as shifting rainfall patterns and temperature extremes introduce major uncertainties in crop yields and food supply. Hybrid GA-ANN models have shown promise in predicting such yield fluctuations more reliably than traditional regression methods, providing valuable foresight for adaptive planning

[95].

The integration of IoT devices and satellite remote sensing into food security monitoring is another emerging direction. These systems generate vast, real-time datasets on soil health, crop growth, and supply logistics. When combined with fuzzy logic and ANN frameworks, they enable proactive monitoring of agricultural performance and early identification of risks such as drought, pest outbreaks, or supply chain disruptions [96]. However, such data sources are often sparse or incomplete. Hybrid GA-ANN approaches, especially when coupled with robust imputation techniques, have demonstrated resilience against missing data, allowing for reliable predictions even under fragmented information [97].

A growing body of research emphasizes the role of intelligent early-warning systems in food security. By combining fuzzy-based reasoning with ANN prediction engines, these systems can detect weak signals of disruption and provide actionable alerts to policymakers, reducing the likelihood of crises escalating unchecked [98]. To achieve this at scale, GA-ANN models have increasingly been embedded into cloud-based infrastructures, enabling distributed processing and real-time analysis across diverse geographic regions [99].

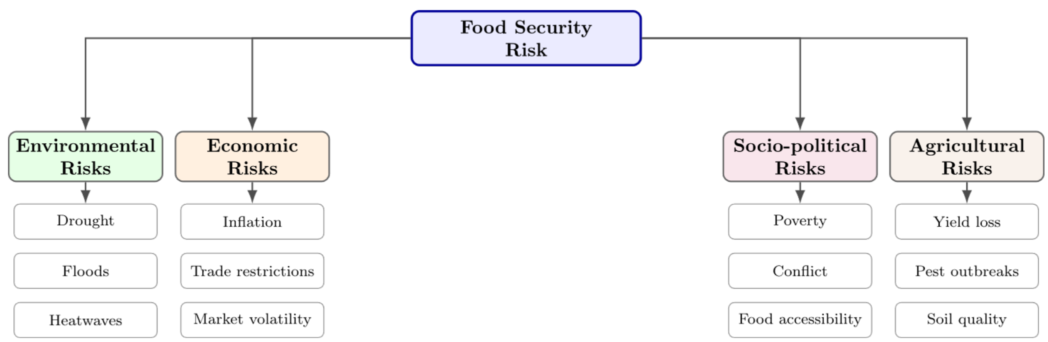

Figure 1: Major Drivers of Food Security Risk

Food security is not only determined by physical availability but also by socio-economic conditions. Indicators such as household income, inflation, and trade restrictions play a pivotal role in accessibility. Fuzzy-based systems are particularly effective at translating ambiguous socio-economic indicators into structured inputs for ANN-driven predictions, enhancing the reliability of food security risk assessments [100]. Moreover, policy interventions such as subsidies or import regulations have long-term consequences for supply-demand balance. Hybrid GA-ANN and FL systems can simulate alternative policy scenarios, supporting more informed and forward-looking decision-making [101]. Food security is shaped by a complex combination of environmental, economic, social, political, and agricultural factors; rather than acting independently, these drivers often interact and reinforce one another, influencing food availability, access, utilization, and stability. To provide a structured overview of these relationships, the following figure summarizes the principal sources of food security risk examined in this review.

Environmental disruptions such as droughts, floods, and climate variability directly affect agricultural productivity, while economic and governance-related factors influence access to food and market functioning. Recognizing these interdependencies is essential for the development of intelligent assessment systems capable of capturing the complexity of real-world food security conditions. Furthermore, the adaptability of computational systems to local contexts remains critical. Food security risks differ substantially between developed and developing countries, as well as across agro-ecological zones. Localized fuzzy inference systems optimized by GA-ANN models have demonstrated effectiveness in smallholder farming contexts, where food security risks are often most acute [102]. Moving forward, the convergence of GA-ANN, FL, and big-data-driven protocols into unified frameworks represents a promising frontier. These integrated systems can simultaneously manage uncertainty, optimize performance, and provide interpretable insights, thus offering a comprehensive foundation for data-driven policy and practice in food security risk management [10].

This review paper seeks to comprehensively evaluate the role of GA-ANN and Fuzzy Logic based systems in food security risk assessment. Drawing from six foundational studies, it identifies their individual strengths, synergy, and applicability in real-world settings. Ultimately, the paper aims to outline a conceptual roadmap for integrating GA-ANN and FL into a unified framework, enhanced by big data protocols, to improve prediction accuracy, resilience, and policy responsiveness in food security management.

2. Research Background

The study of food security risk assessment has evolved alongside advances in artificial intelligence, particularly in fields that require modeling of complex, nonlinear, and uncertain systems. Traditionally, risk assessments in food systems relied heavily on deterministic or semiquantitative models that incorporated statistical thresholds and expert judgment to define vulnerability, exposure, and impact across supply chains [8]. However, with the growing volume, velocity, and variety of agricultural and environmental data, conventional models have shown limitations in scalability, adaptability, and interpretability.

Artificial Neural Networks (ANNs) have emerged as robust tools for learning complex patterns from large, high-dimensional datasets. In agriculture, they have been extensively used for yield prediction, soil analysis, irrigation modeling, and post-harvest quality control [2]. Despite their effectiveness, ANNs are often challenged by issues of slow convergence, local minima entrapment, and sensitivity to initialization parameters. To mitigate these drawbacks, researchers have increasingly employed Genetic Algorithms (GA) to optimize the structure and weight parameters of ANN models, resulting in GA-ANN hybrids that outperform standalone neural networks in both accuracy and generalization [3], [1].

In parallel, Fuzzy Logic (FL) systems have gained attention for their ability to model human-like reasoning under uncertainty. Unlike binary logic systems, FL operates on degrees of truth, which makes it suitable for handling linguistic variables (e.g., "low yield", "moderate risk") and vague data a common characteristic in food security domains where data may be imprecise, incomplete, or subjective [4]. Fuzzy logic has been employed to assess food security at both micro (crop-level) and macro (national policy) levels, where economic indicators, production data, and climate conditions interact in unpredictable ways [5].

The convergence of these techniques has led to the development of integrated models such as Genetic Neuro-Fuzzy (GNF) systems, which combine the learning capability of ANNs, the global optimization ability of GA, and the uncertainty modeling strength of FL. These systems aim to support real-time decision-making in dynamic environments by producing interpretable outputs with high prediction accuracy [1]. For instance, in a notable application, a fuzzy logic-based framework was used to classify food security risk using a dataset of over 500,000 driving journeys as a proxy for economic activity and distribution logistics, demonstrating an accuracy of 90% across pessimistic, neutral, and optimistic scenarios [6].

On a national scale, the application of backpropagation (BP) neural networks for early warning systems has been explored in China, where food security risks were classified into multiple levels (no warning, light warning, moderate warning, and heavy warning) based on indicators such as grain output, ecological conditions, and trade dynamics [6]. The model combined principal component analysis (PCA) and analytic hierarchy process (AHP) to construct a multidimensional risk index, validated against data from 2000 to 2019.

Additionally, fusion models that integrate unsupervised clustering, expert knowledge weighting such as AHP, DEMATEL and ensemble machine learning have shown promising results in domain-specific assessments such as rice production risk across 31 provinces in China. These models can handle high-dimensional hazard detection data and yield classification accuracy exceeding 99%, validating their reliability for regulatory and policy applications [5].

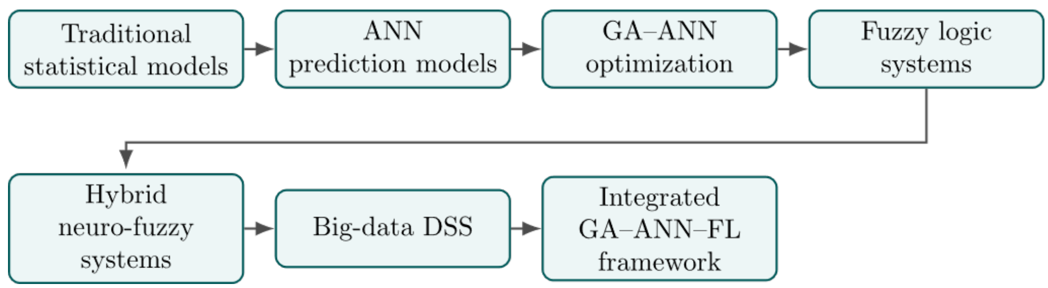

Figure 2: Evolution of AI-Based Food Security Assessment Approaches

The combination of GA-ANN and FL techniques within food security modeling is not merely additive it is synergistic. GA improves the adaptability of ANN, while FL adds interpretability and robustness in the face of ambiguous inputs. Together, they provide a foundation for the next generation food security risk assessment systems that are both precise and explainable. This integrated approach not only addresses prediction tasks but also supports strategic decisions for early warning systems, sustainable planning, and supply chain optimization. Research on food security assessment has evolved considerably over the past decades. Early studies relied primarily on statistical and econometric techniques, whereas more recent work has incorporated machine learning, optimization methods, and hybrid intelligent systems. Figure 2 outlines the main stages of this methodological evolution.

3. Related Studies

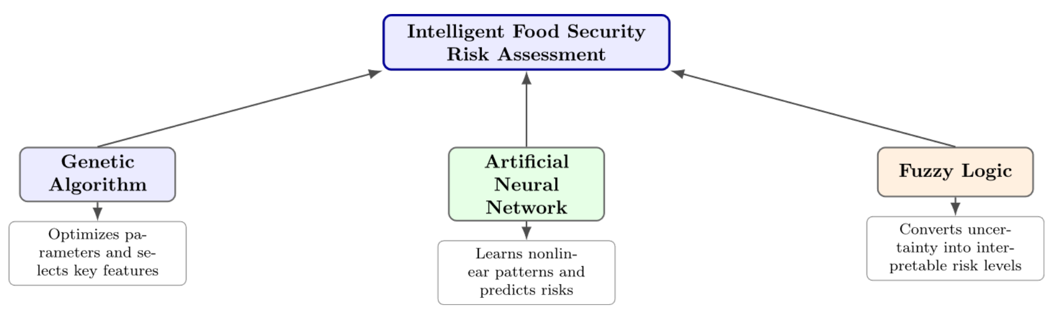

Figure 3: Functional Roles of GA, ANN and Fuzzy Logic in Food Security Assessment

Research on food security risk assessment has advanced significantly in recent years, driven by the integration of machine learning, fuzzy logic, hybrid soft computing, and big-data-driven decision support systems. The following review clusters relevant studies into four main thematic areas: neural-based approaches, fuzzy and hybrid methods, big-data-driven decision support, and early warning/ecological frameworks. The intelligent approaches reviewed in this study rely on several complementary computational techniques. Before reviewing individual contributions, it is useful to distinguish the specific role that Genetic Algorithms, Artificial Neural Networks, and Fuzzy Logic typically play within food security applications.

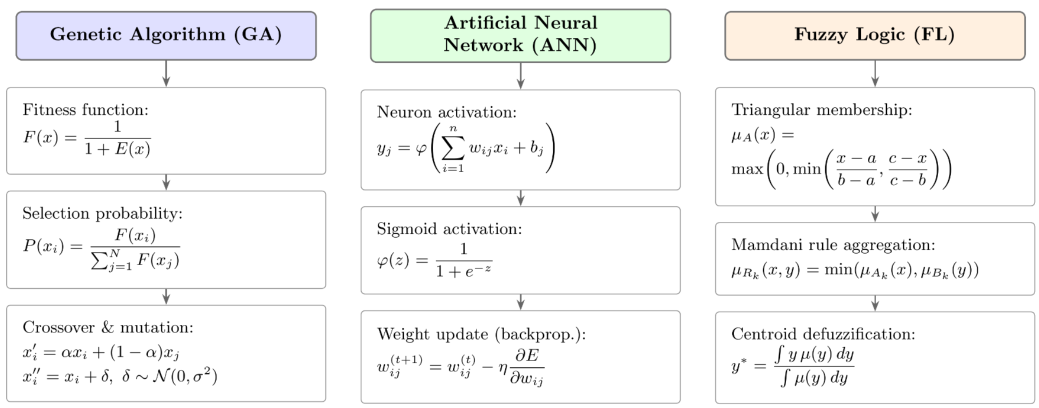

Figure 4: Mathematical Formulation of GA, ANN, and Fuzzy Logic Components

Figure 4: Mathematical Formulation of GA, ANN, and Fuzzy Logic Components

3.1. Neural-based Approaches

Artificial neural networks and related machine-learning methods have been widely applied to food security risk assessment. The study in [12] proposed to fuse multiple machine-learning models for rice security assessment, achieving improved detection accuracy across heterogeneous datasets. Similarly, [13] proposed to optimize banana yield in greenhouse environments through a combination of generalized regression neural networks (GRNN) and genetic algorithms for nutrient optimization. Predictive modeling approaches have also been extended to food safety, such as [7], which applied tree-based ML algorithms to predict *Salmonella* outbreaks, and [32] and [33], which forecast acute food insecurity by integrating high-frequency health and environmental indicators. Ensemble methods such as Bayesian model averaging [19] and hybrid random forest–MLP models [27] have been proposed to enhance ecological security evaluations in grain-producing areas. In addition, [20] introduced a hybrid GA–fuzzy neural model tailored for crop yield prediction, highlighting the potential of combining neural inference with evolutionary optimization. Similarly, [21] demonstrated the application of ANN-based frameworks for early detection of food insecurity at regional scales, reinforcing the value of neural approaches in anticipatory risk management.

More recent efforts extend neural-based forecasting to global markets [51] applied recurrent neural networks to predict staple food price volatility under climate stress, while [52] used convolutional neural networks (CNNs) to classify crop stress signals from satellite imagery, improving early identification of yield losses. Likewise, [53] developed a deep reinforcement learning framework to optimize irrigation decisions under uncertain rainfall, demonstrating the adaptability of neural models to decision-oriented contexts. Across these studies, neural-based approaches demonstrate high predictive accuracy, but they often face challenges in interpretability

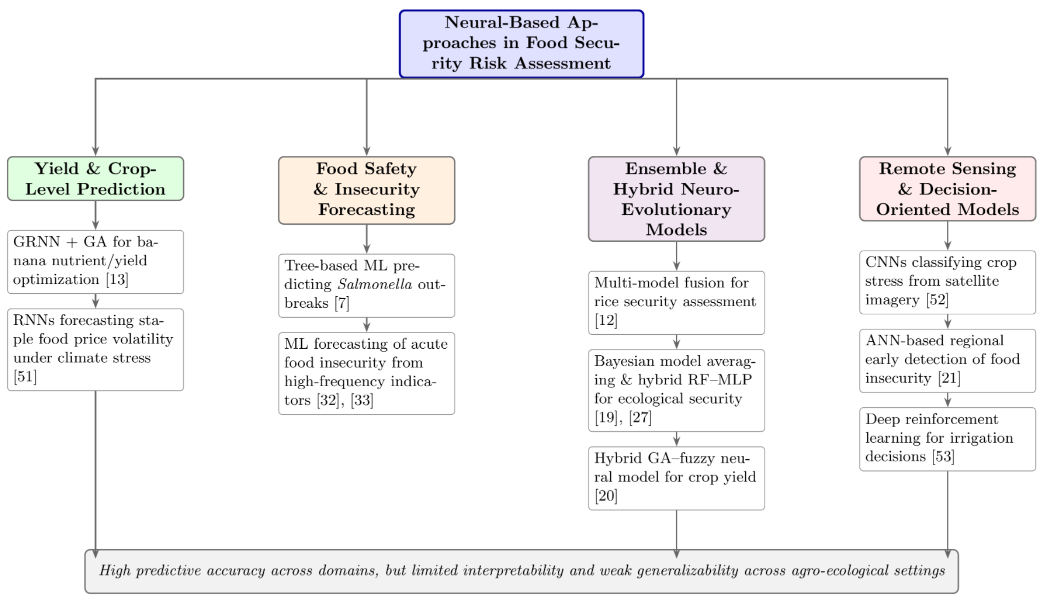

Figure 5: Taxonomy of Neural-Based Approaches in Food Security Risk Assessment

and generalizability across diverse agro-ecological settings.

3.2. Fuzzy and Hybrid Soft-Computing

Fuzzy systems and hybrid models combining fuzzy logic with neural networks or evolutionary algorithms have been applied to address uncertainty in food security risk modeling. The study [15] proposed a Mamdani fuzzy logic system for assessing food security risk levels, while [14] introduced a hybrid fuzzy-neural approach to reduce uncertainty in agricultural predictions. Hybrid optimization techniques such as ANFIS combined with multi-objective genetic algorithms [22, 26] have been proposed for yield forecasting, while fuzzy-based DSS frameworks [25, 36] enable resilient decision making under imprecise information. Applications in agricultural supply chains have been a key focus [37] and [38] applied fuzzy TOPSIS and fuzzy-AHP methods to evaluate risks in financing and rice supply chains, respectively, while [40] used spherical fuzzy AHP to model pandemic-driven disruptions. Other works, such as [39] and [46], demonstrated hybrid fuzzy methods for supply chain performance evaluation, integrating risk metrics like time, cost, and quality. Further, [54] proposed a Z-number fuzzy inference system for food logistics under uncertainty, and [55] presented a hybrid GA–fuzzy model to optimize fertilizer use efficiency across diverse crop systems. Recent innovations also include [34] and [35], which applied interval type-2 fuzzy logic and intuitionistic fuzzy models, respectively, to agricultural decision-making under high uncertainty, demonstrating that extended fuzzy frameworks can better handle ambiguous and conflicting inputs in food system risk scenarios. These approaches highlight the adaptability of fuzzy systems for modeling imprecision, but they also require expert input, which can introduce subjectivity in risk assessment.

3.3. Big-Data Driven Decision Support Systems

The rise of big data has enabled the construction of advanced decision support systems (DSS) for food security. The study in [17] proposed an autonomous fuzzy DSS that integrates multisource big data for risk assessment, while [18] surveyed Agriculture 4.0 DSS, emphasizing MLenabled farm-level decision support. Studies such as [7, 43, 44] have focused on integrating large heterogeneous datasets, including news text, climate indicators, and conflict event data, to predict food insecurity. Data harmonization efforts such as the HFID dataset [42] and explainable AI frameworks [24] demonstrate progress toward transparent and scalable monitoring tools. Meanwhile, organizations such as WFP have explored machine-learningdriven early warning platforms [45], while [41] examined socio-economic drivers of food insecurity using ML on survey data. Additional advances include [56], which proposed federated learning architectures for cross-country food risk analysis, and [57], which integrated graph neural networks with big-data pipelines to map cascading food supply disruptions. Similarly, [58] developed a blockchain-enabled DSS for agricultural trade risk monitoring, showing the potential for coupling data integrity with AI-driven prediction. Together, these studies underline the potential of big-data DSS in providing real-time, data-driven insights for proactive interventions.

3.4. Early Warning Systems and Ecological Risk Frameworks

Figure 6: Classification of Reviewed Studies

Early warning systems (EWS) are critical for mitigating food security risks before crises escalate. Neural and hybrid models have been widely adopted in this space [16] used a BP neural network combined with AHP for China's food security early warning, while [28] proposed a 3D visualization platform integrating GA-BP models for dynamic monitoring. Similarly, [23] and [29] improved BP and anomaly detection methods for food safety surveillance. Ecological and supply-chain risk frameworks complement these EWS. Studies such as [30] and [48] assessed ecological security in grain-producing regions, while [47] applied fuzzy logic to supply chain risk in perishable foods. Fuzzy FMEA models [49] and real-time event-driven fuzzy supply chain monitoring [50] illustrate the potential for combining classical risk management frameworks with modern data streams. More recently, [59] applied ensemble anomaly detection for locust outbreak monitoring, while [60] integrated satellite-derived drought indices with fuzzy EWS models for sub-Saharan Africa. These works demonstrate strong progress in EWS, but challenges remain in ensuring scalability, cross-regional applicability, and integration of socioeconomic indicators alongside biophysical data. To support a systematic analysis of the literature, the selected studies were organized into four thematic categories according to their methodological orientation and primary application focus.

3.5. Summary

Overall, neural-based models excel at prediction accuracy but often lack transparency; fuzzy and hybrid systems manage uncertainty effectively but can be resource-intensive; big-data DSS platforms offer scale and real-time monitoring but face integration challenges; and early warning systems combine these approaches but require more cross-disciplinary data sources. Future research should focus on hybrid frameworks that balance predictive power, interpretability, and practical usability for policy and on-farm decision making.

4. Research Challenge

Despite significant progress in applying artificial intelligence (AI), machine learning (ML), and fuzzy-based approaches to food security risk assessment, there are still substantial gaps that hinder their large-scale and practical deployment. This section highlights ten key challenges, each of which requires further investigation to move from theoretical contributions to actionable, policy-driven solutions.

4.1. Data Quality and Availability

The foundation of any AI or ML model is data. However, the availability of reliable, high-quality datasets remains a significant obstacle. Agricultural and food security data are often fragmented, outdated, or collected inconsistently across regions. In rice security assessment, for example, accurate predictions depend on integrating climatic, soil, and socio-economic variables, yet such comprehensive datasets are rarely available in low-resource countries [61]. The lack of standardization in data collection further complicates model training and validation, often resulting in biased or underperforming models when applied outside of their initial test environment.

4.2. Data Integration Across Sources

Beyond data availability, the challenge of combining multiple data sources into a coherent framework persists. Food security is influenced by a wide array of factors, ranging from crop yield and weather conditions to market fluctuations and political stability. Greenhouse banana yield prediction, for instance, requires the simultaneous collection of nutrients (nitrogen, potassium, magnesium) and environmental data [62]. Integrating such heterogeneous data streams into a unified model is complex, as it demands advanced data fusion techniques that can handle temporal, spatial, and contextual variability.

4.3. Model Complexity vs. Interpretability

One of the paradoxes in food security risk modeling is the trade-off between accuracy and interpretability. Hybrid models combining genetic algorithms, neural networks, and fuzzy logic can produce highly accurate results [9], [4]. However, their complexity makes them difficult for non-technical stakeholders, such as policymakers or farmers, to understand. The "black box" nature of neural networks limits their transparency, raising concerns about accountability in decision-making. For food security to benefit from AI, researchers must develop models that are both accurate and interpretable, possibly through explainable AI techniques.

4.4. Scalability for Big Data

Scalability remains a technical bottleneck. Many AI-driven systems are tested on relatively small datasets or in controlled environments, but they falter when applied to national or global data flows. With the rise of big data sources such as remote sensing imagery, IoT-based farm sensors, and large agricultural databases models must be capable of handling vast amounts of information in near real-time [11]. Unfortunately, most current models cannot efficiently process such volumes without sacrificing performance or accuracy, limiting their applicability in operational food security monitoring systems.

4.5. Real-Time Decision Support

Timely intervention is critical in food security, especially when responding to emerging threats such as droughts, pest outbreaks, or sudden market shifts. However, many existing fuzzy decision support systems and neural network frameworks suffer from processing delays [63]. This latency reduces their usefulness in contexts where rapid responses are required to mitigate food crises. Building lightweight, real-time algorithms that can provide instant risk assessments remains an open challenge, especially in resource-constrained settings where computational infrastructure is limited.

4.6. Regional Adaptability and Transferability

Most models are designed and tested within specific geographic or socio-economic contexts. For example, BP neural networks trained on Chinese datasets for early warning systems [11] may not generalize well to African or Latin American contexts, where agricultural practices, climate patterns, and market dynamics differ significantly. Adapting models to new regions often requires retraining with local data, which may not be available. Developing adaptable and transferable frameworks that can learn across domains with minimal reconfiguration would enhance the global relevance of food security risk models.

4.7. Policy Integration and Usability

Another major limitation lies in the gap between technical advances and governance. Policymakers often find AI-driven tools too complex or misaligned with policy frameworks [4], [63]. Even when models identify risks accurately, their outputs may not translate into actionable recommendations. For instance, fuzzy models may classify risk levels but fail to suggest concrete interventions. Building user-friendly interfaces and decision support systems that bridge the technical-policy gap is crucial to ensure that computational tools contribute meaningfully to governance and resource allocation.

4.8. Uncertainty Management

Food systems are inherently uncertain, influenced by unpredictable factors such as climate change, global trade disruptions, and social unrest. Fuzzy logic has been widely used to capture this uncertainty [9], [4], yet its capacity is often limited when dealing with highly dynamic, multifactor interactions. Moreover, uncertainty is rarely communicated effectively to policymakers or farmers, who need to understand the level of confidence associated with each prediction. Improving methods to quantify, propagate, and explain uncertainty in food security assessments is therefore essential to support informed, risk-aware decision-making.

4.9. *Balancing Accuracy with Computational Efficiency*

High-accuracy models such as deep neural networks often come at the cost of computational intensity. These models require advanced hardware, large-scale storage, and high energy consumption. In developing countries where food security risks are most acute such resources are frequently unavailable [62], [63]. This creates a paradox: the most vulnerable regions may not be able to implement the very systems designed to protect them. Research must therefore focus on developing lightweight models that balance accuracy with efficiency, making them accessible and sustainable across diverse contexts.

4.10. *Ethical and Social Acceptance*

Ethical and social issues must not be overlooked. The use of AI in food security raises questions about data privacy, fairness, and accountability. For example, if models are trained on biased data, they may reinforce existing inequalities by prioritizing certain crops or regions [61], [4]. Additionally, trust is a critical factor: stakeholders may resist adopting AI-driven systems if they fear loss of control or lack confidence in technology. Addressing these ethical and social challenges is necessary to ensure that AI adoption in food security is both inclusive and equitable.

4.11. *Climate Change Impacts on Model Reliability*

Climate change is one of the greatest threats to global food security because it introduces significant variability and unpredictability into agricultural systems. Traditional AI and ML models are often trained on historical datasets that assume relatively stable climate conditions.

However, extreme weather events such as floods, droughts, hurricanes, and heatwaves are becoming more frequent and intense due to climate change [84]. These anomalies disrupt crop yields, supply chains, and market stability, rendering past trends a poor predictor of future risks. For example, a model trained on past rice yield and rainfall data may significantly underestimate losses caused by a sudden prolonged drought. Furthermore, most food security datasets underrepresent these extremes, which leads to biased models that fail in crisis conditions. To improve resilience, researchers must integrate climate projection models, remote sensing of environmental stressors, and scenario-based simulations into food security assessments. This would allow models to better account for non-linear, climate-driven disruptions and improve their reliability in forecasting under uncertain environmental futures.

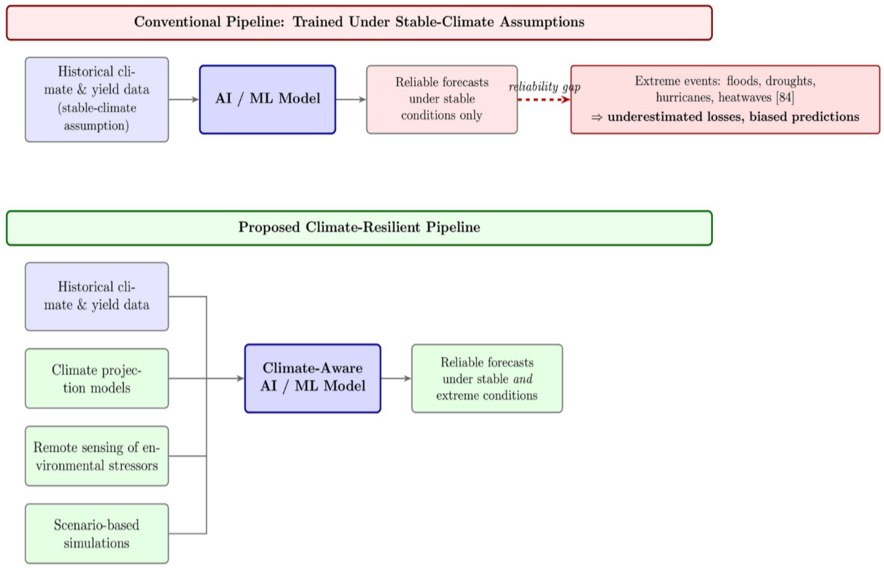

Figure 7: Climate Change Impacts on Model Reliability

Figure 7 contrasts a conventional forecasting pipeline, trained under the assumption of stable climate conditions, with the climate-resilient pipeline proposed in this review. Because extreme events such as floods, droughts, hurricanes, and heatwaves are underrepresented in historical training data [84], conventional models develop a growing reliability gap and systematically underestimate losses during crisis conditions. Integrating climate projection models, remote sensing of environmental stressors, and scenario-based simulations alongside historical data allows the resulting climate-aware model to remain reliable under both stable and extreme conditions.

4.12. Socio-Economic Factors in Food Security Models

While technological and environmental factors are central to food production, food security also depends heavily on socio-economic determinants. Issues such as household income, infrastructure, access to markets, food prices, and distribution networks can significantly influence food availability and accessibility [85]. Many AI-driven models disproportionately focus on biophysical variables like soil quality, temperature, or rainfall while neglecting socioeconomic dimensions. For instance, even when agricultural yields are high, vulnerable populations may still face food insecurity if poverty prevents them from purchasing adequate food. Similarly, conflict, political instability, or trade restrictions can cause severe disruptions that models based purely on agricultural production cannot capture. Incorporating socio-economic datasets such as household surveys, trade indices, and market price fluctuations would make predictive systems more holistic and policy relevant. This integration would allow researchers to move beyond yield prediction toward comprehensive food system risk assessments that address both availability and accessibility.

4.13. Cross-Disciplinary Collaboration Barriers

Food security is a complex challenge that spans multiple disciplines, including agricultural sciences, economics, data science, climatology, and policy studies. However, much of the current research is siloed, with computer scientists developing advanced algorithms in isolation from agricultural experts or social scientists [86]. This lack of collaboration often produces technically sophisticated models that may not align with the realities of farming systems or the needs of policymakers. For example, a highly accurate neural network model predicting crop yields may not be actionable for farmers if it fails to consider practical issues such as irrigation practices, pest management, or supply chain constraints. Similarly, policymakers may find outputs too technical or abstract to integrate into governance frameworks. Bridging these disciplinary divides requires building interdisciplinary teams, fostering cross-sector partnerships, and creating collaborative platforms where expertise from multiple domains can converge. Only through such collaboration can AI-driven models evolve into practical, context-aware tools that support sustainable food security strategies.

4.14. Limited Explainability in Fuzzy-Logic–Based Models

Fuzzy logic has been widely promoted to model uncertainty in food security because it mimics human reasoning and can provide interpretable rule-based outputs. However, as fuzzy systems scale up and include hundreds or even thousands of fuzzy rules, they themselves can become opaque and difficult to interpret [87]. For instance, a fuzzy decision support system for food insecurity may rely on overlapping rules covering climatic, economic, and social dimensions, but the interaction among these rules can create outputs that are difficult for stakeholders to trace back to specific factors. Policymakers and farmers often require simple, actionable insights, not overwhelming rule sets. If outputs are too complex, the intended transparency of fuzzy logic may be lost, undermining trust and usability. Future research should focus on developing methods for simplifying fuzzy models, visualizing outputs in user-friendly formats, and integrating explainable AI (XAI) techniques that preserve interpretability without sacrificing accuracy.

4.15. Temporal Data Gaps and Forecasting Limitations

Food security risk assessment requires not only cross-sectional data but also robust temporal datasets that track changes over time. Unfortunately, many agricultural datasets are either collected annually or irregularly, which severely limits forecasting capacity [88]. This is particularly problematic for early warning systems, which depend on near-real-time data to predict seasonal risks like droughts, pest outbreaks, or price shocks. For example, relying solely on annual yield statistics does not capture mid-season crop stress caused by erratic rainfall or pest infestations. Temporal gaps also limit the ability of machine learning models to detect patterns or trends in food security risks, resulting in forecasts that may be outdated or incomplete. Strengthening data collection through high-frequency monitoring technologies such as IoT-based farm sensors, mobile surveys, and remote sensing would provide the temporal granularity necessary for dynamic forecasting. This would enable proactive responses rather than reactive crisis management.

4.16. Generalization Across Crop Types

Most AI and ML models developed for food security are highly crop-specific, trained and validated on datasets for staple crops such as rice, maize, or wheat [89]. While these models may perform well within a given crop system, their predictive capacity often fails when applied to other crops with different physiological and ecological characteristics. For instance, a model optimized for rice yield prediction may not generalize to banana or cassava due to differences in nutrient uptake, disease vulnerability, and growing conditions. This crop-specific limitation reduces the utility of AI-based systems in regions with diverse agricultural systems. Developing multi-crop models or creating transferable frameworks that can adapt across crop types with minimal reconfiguration would increase the versatility and scalability of food security risk assessments. Such models would also support policy frameworks that need to account for the full spectrum of agricultural production in each region.

4.17. Integration with Remote Sensing Technologies

Remote sensing technologies, particularly satellite imagery, offer a powerful means of monitoring agricultural systems at scale, providing critical data on vegetation health, soil moisture, and land use [90]. However, integrating this vast and complex data into AI-driven food security assessments remains underdeveloped. Challenges include processing high-resolution imagery, aligning satellite data with ground-truth observations, and handling noise caused by atmospheric conditions or cloud cover. Moreover, most food security models have yet to fully exploit the temporal richness of remote sensing data, which can provide near-real-time monitoring of crop conditions. Improved integration of remote sensing with machine learning models would enhance spatial and temporal coverage of food security risk assessments, enabling early detection of stressors and more targeted interventions. This integration could transform food security monitoring from a largely reactive process into a proactive, data-driven system.

4.18. Resilience to Data Noise and Incompleteness

Agricultural and food security datasets are frequently noisy, incomplete, or inconsistent. Missing values may occur due to logistical challenges in data collection, while noise can arise from errors in manual surveys, faulty sensors, or environmental variability [91]. Many machine learning algorithms, particularly deep learning models, are highly sensitive to such imperfections and may produce unreliable predictions when applied to noisy data. This is a major limitation in developing countries where data collection infrastructure is weak. Current solutions such as data imputation or preprocessing can help, but they often introduce additional biases. Research should focus on building robust models that are resilient to imperfect data, capable of learning from incomplete information, and able to quantify uncertainty arising from noise. Such resilience is critical for ensuring that food security risk assessments remain reliable in real-world, resource-limited contexts.

4.19. Lack of Benchmark Datasets and Evaluation Standards

Unlike fields such as computer vision or natural language processing, where benchmark datasets like ImageNet or GLUE exist, food security research lacks standardized datasets and evaluation frameworks [92]. This absence makes it difficult to compare model performance across studies or replicate findings, slowing down scientific progress. For instance, one study may evaluate food security risk using yield and rainfall data, while another uses socio-economic indicators, making direct comparisons nearly impossible. Establishing benchmark datasets with standardized features and evaluation protocols would foster transparency, reproducibility, and innovation in the field. Such benchmarks would also create a common ground for interdisciplinary collaboration, helping researchers align their efforts toward shared objectives.

4.20. Adoption Barriers in Low-Income Countries

Figure 8: Taxonomy of research challenges in food security risk assessment, categorized by data, model, deployment, and socio-economic dimensions

Even when advanced AI-based systems for food security risk assessment exist, their adoption in low-income countries remains limited due to a variety of barriers [93]. These include weak digital infrastructure, low computational capacity, lack of technical expertise, and institutional resistance to change. Furthermore, vulnerable populations often have low levels of digital literacy, limiting their ability to use AI-driven decision support systems. This creates a paradox the region most vulnerable to food insecurity is often the least equipped to benefit from cutting-edge solutions. Overcoming these adoption barriers requires not only developing lightweight, low-cost technologies but also investing in capacity building, institutional partnerships, and inclusive governance frameworks. Research should therefore shift toward implementation strategies that are tailored to local contexts, ensuring that technological innovations are both accessible and sustainable. Despite substantial advances in intelligent food security assessment, a few challenges continue to hinder the effectiveness and large-scale adoption of existing systems. These challenges extend beyond algorithmic performance and involve broader technical, organizational, and societal considerations.

4.21. Summary Table of Research Challenges

Table 1: Summary of key research challenges in AI/ML-based food security risk assessment.

Challenge	Ref.	Description
1. Data Quality and Availability	[61]	Lack of reliable, standardized, and comprehensive datasets; fragmented or outdated data hinders robust model development.
2. Data Integration Across Sources	[62]	Difficulty in merging heterogeneous sources (climate, soil, socio-economic, IoT, satellite) into a coherent analytical framework.
3. Model Complexity vs. Interpretability	[9, 4]	Hybrid AI models achieve accuracy but act as "Black boxes," limiting transparency and trust for decision-makers.
4. Scalability for Big Data	[11]	Many AI systems fail to handle large-scale datasets (IoT, remote sensing) efficiently at national or global levels.
5. Real-Time Decision Support	[63]	Existing models suffer from latency, limiting timely interventions during sudden food crises.
6. Regional Adaptability and Transferability	[11]	Models trained in specific regions often fail elsewhere; retraining requires local data that may not be available.
7. Policy Integration and Usability	[4, 63]	Outputs are often too technical, lacking actionable insights for policymakers and resource managers.
8. Uncertainty Management	[9, 4]	Current fuzzy and AI methods capture uncertainty only partially; dynamic multi-factor uncertainty is poorly managed.
9. Balancing Accuracy with Computational Efficiency	[62, 63]	High-accuracy models demand advanced infrastructure, which is often unavailable in vulnerable regions.
10. Ethical and Social Acceptance	[61, 4]	Issues of data privacy, fairness, and trust reduce willingness of stakeholders to adopt AI-driven systems.

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Table 1 – *continued from previous page*

Challenge	Ref.	Description
11. Climate Change Impacts on Model Reliability	[84]	AI models trained in historical data struggle to account for climate-driven anomalies like droughts and floods.
12. Socio-Economic Factors in Food Security Models	[85]	Many models prioritize biophysical factors while neglecting socio-economic drivers such as poverty and trade access.
13. Cross-Disciplinary Collaboration Barriers	[86]	Limited collaboration between computer scientists, agricultural experts, and policymakers reduces real-world usability.
14. Limited Explainability in Fuzzy Logic Models	[87]	Large fuzzy systems with complex rule sets become opaque, undermining their intended transparency.
15. Temporal Data Gaps and Forecasting Limitations	[88]	Infrequent or annual data collection limits dynamic forecasting for seasonal risks and early warning systems.
16. Generalization Across Crop Types	[89]	Most models are crop-specific, reducing their adaptability to diverse agricultural systems.
17. Integrations with Remote Sensing Technologies	[90]	Difficulty in incorporating high-resolution satellite imagery into AI models due to noise and alignment issues.
18. Resilience to Data Noise and Incompleteness	[91]	Many AI systems are sensitive to missing or noisy agricultural data, reducing reliability in low resource contexts.
19. Lack of Benchmark Datasets and Evaluation Standards	[92]	Absence of standard datasets and evaluation protocols hinders reproducibility and model comparison.
20. Adoption Barriers in Low-Income Countries	[93]	Weak digital infrastructure, limited technical skills, and institutional barriers slow adoption of AI solutions.

5. Future Research Directions

The challenges identified throughout literature suggest several priorities for future research. There is a growing need for assessment systems that are more integrated, transparent, scalable, and operationally effective.

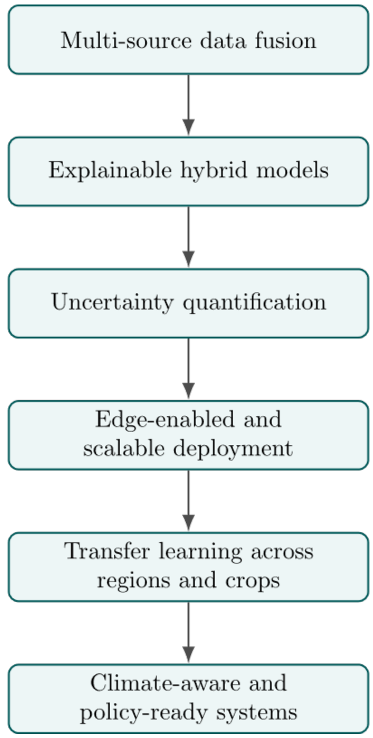

Figure 9: Future Research Roadmap

5.1. Advanced Multi-Source Data Fusion

Food security risk assessment requires data that is often heterogeneous, incomplete, or inconsistent across sources. Traditional approaches rely on limited datasets, leading to gaps in predictive accuracy and coverage. Future work should focus on fusing multi-source datasets such as remote sensing imagery, IoT-based soil and weather sensors, agricultural management records, market transactions, and even social media trends to build a more holistic risk assessment framework. Ensemble-based fusion methods have already shown promise in rice production [64], and similar architectures can be extended to other crops and broader geographic regions. Advanced techniques such as graph neural networks, multimodal transformers, and federated learning may further strengthen cross-domain integration while handling data privacy concerns. For example, integrating market and climate data with yield forecasts could enable early warnings of supply disruptions. A major challenge is the asynchronous nature of these datasets, where temporal alignment is difficult, and missing values are common. Future studies should therefore design robust fusion models that can learn from sparse, noisy, or delayed inputs, ensuring reliable predictions for early-warning systems [64], [68].

5.2. Interpretable Hybrid Models and Explainability-by-Design

Many of the hybrid methods explored for agricultural prediction and risk analysis such as neural networks combined with genetic algorithms or fuzzy logic offer strong performance but remain opaque to non-technical users [65], [66]. For decision-making in food security, interpretability is as important as accuracy, since stakeholders such as policymakers and farmers need to understand why a system is signaling a risk or suggesting a course of action. Future research should embed explainability into hybrid models by developing rule-based fuzzy systems that translate neural outputs into human-readable logic. Attention mechanisms, feature-importance metrics, and causal inference frameworks can be incorporated to highlight the drivers of food insecurity (e.g., fertilizer levels, weather fluctuations). Such explainable-by-design frameworks can also help detect biases in training data and guide policy interventions more effectively. Another promising direction is integrating interactive visualization dashboards that present model outputs alongside explanations, allowing experts to challenge or validate the system's reasoning. By enhancing trust and transparency, interpretable hybrid models will increase the likelihood of adoption by governments and agricultural agencies [65], [66].

5.3. Uncertainty Quantification and Probabilistic Risk Outputs

Most food security risk models provide deterministic predictions, which can be misleading under volatile conditions such as climate extremes or pest outbreaks. Future research should focus on developing probabilistic models that explicitly quantify uncertainty in predictions. For example, extending fuzzy logic frameworks with Bayesian methods or Monte Carlo simulations can generate confidence intervals around risk scores [83], [68]. This will allow decision-makers to weigh risks under uncertainty and prioritize interventions based on both severity and confidence levels. In early-warning systems, calibrated uncertainty can reduce the rate of false alarms that might otherwise erode trust in automated tools. Recent progress in uncertainty-aware deep learning such as Bayesian neural networks and ensemble-based variance estimation can be adapted for crop yield prediction and market volatility assessment. Another promising avenue is probabilistic graphical models that capture causal dependencies between weather patterns, soil conditions, and crop yields. By explicitly modeling uncertainty, researchers can build systems that are not only accurate but also more reliable for high-stakes decision-making [83], [68].

The convergence of optimization algorithms, machine learning, uncertainty modeling, and big-data infrastructures motivates the development of a unified food security assessment framework.

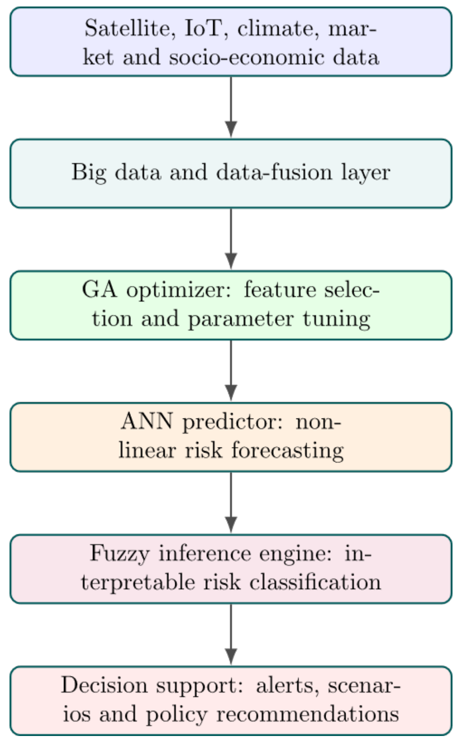

Figure 10: Proposed Integrated GA-ANN-Fuzzy Architecture for Food Security Risk Assessment

5.4. Real-Time, Edge-Enabled, and Scalable Deployments

While many advanced algorithms demonstrate success in controlled experiments, their deployment in real agricultural systems remains limited due to resource constraints. Farmers and local agencies often lack access to high-performance computing infrastructure or stable internet connectivity, which makes cloud-based solutions impractical. Future research should focus on designing lightweight, compressed versions of hybrid models that can run on edge devices such as smartphones or low-power IoT sensors [64], [68]. Techniques like model pruning, quantization, and federated learning can reduce computational demands without significantly compromising accuracy. Real-time adaptability is another critical challenge: models must be capable of incremental learning from streaming sensors or satellite data rather than relying on static retraining cycles. Additionally, scalability is crucial when expanding from pilot studies to national or regional early-warning systems. Researchers should therefore investigate distributed architectures that balance local edge inference with central coordination for largescale monitoring. This direction will accelerate the practical translation of food security models into on-the-ground applications [64], [68].

5.5. Transfer Learning and Domain Adaptation Across Regions and Crops

One persistent limitation in food security research is the difficulty of generalizing models across diverse geographic, climatic, and crop conditions. Models trained on one dataset (e.g., rice in East Asia) often fail when applied elsewhere (e.g., bananas in tropical greenhouses) due to domain shifts [64], [65], [67]. Future work should explore transfer learning techniques that reuse representations learned in data-rich settings and adapt them to new contexts with minimal labeled data. For instance, a model trained on rice yield prediction could be fine-tuned for wheat or maize with limited new data. Domain adaptation methods such as adversarial learning or instance reweighting could mitigate biases introduced by differences in soil, weather, or cultivation practices. Meta-learning approaches can further accelerate adaptation by training models to rapidly learn from small amounts of new data. Cross-regional collaboration to build shared, open datasets will also be vital. Ultimately, enabling robust generalization will ensure that risk-assessment systems are not restricted to isolated pilot projects but can scale globally [64], [65], [67].

5.6. Coupling Climate Extremes and Crop-Physiology Models with ML

Food production is increasingly vulnerable to climate extremes such as droughts, heat waves, and floods. Traditional machine learning (ML) methods trained solely in historical data often fail to extrapolate under unprecedented conditions. To address this gap, future research should integrate crop-physiology models and climate projections with data-driven ML approaches [65], [67]. For example, physics-informed neural networks can embed agronomic knowledge such as nutrient uptake rates, evapotranspiration, and pest resistance directly into model architectures. By coupling process-based models with ML, researchers can simulate crop behavior under rare or extreme weather scenarios where observational data are scarce. This hybrid framework can also improve the interpretability of predictions by linking risk assessments to underlying physiological and environmental processes. Another important research direction involves incorporating downscaled climate projections from global circulation models into early warning systems, enabling policymakers to anticipate food security risks years in advance. By aligning predictive analytics with agricultural science and climate modeling, this approach can produce robust, forward-looking risk assessments that remain reliable under future climate uncertainty [65], [67].

5.7. Standardized Benchmarks, Datasets, and Evaluation Protocols

A major barrier to progress in food security risk assessment is the lack of standardized datasets and evaluation frameworks. Most studies rely on local, proprietary datasets that are difficult to access or replicate, making cross-comparison of methods challenging [64], [66]. Future research should focus on developing large-scale, open-access, multi-modal datasets that combine climate data, soil quality indicators, yield statistics, and socio-economic variables. Equally important is the need for consistent evaluation protocols, including train/test splits, performance metrics, and baselines. Without such benchmarks, it is difficult to determine whether improvements come from better models or from differences in datasets. Another promising direction is the creation of food-security "challenge competitions" or shared tasks, modeled after initiatives in computer vision and natural language processing. These competitions could encourage reproducibility and accelerate progress by providing standardized testbeds. In addition, evaluation should go beyond accuracy to include fairness, robustness, and interpretability metrics to ensure models are not only predictive but also equitable and trustworthy [64], [66].

5.8. End-to-End Early-Warning Systems and Human-in-the-Loop Decision Support

While many studies focus on building predictive models, fewer address the integration of these models into operational early-warning systems. Future research should aim at designing end-toend pipelines that cover the entire workflow from data acquisition and preprocessing to prediction, visualization, and policy communication [67], [68]. Human-in-the-loop approaches are particularly important in this context. Automated systems can provide risk alerts, but expert intervention is often necessary to contextualize results, verify anomalies, and fine-tune model thresholds. Research should explore interactive interfaces that allow experts to annotate outputs, provide corrections, and guide model updates. Such feedback loops not only improve system accuracy but also enhance trust and adoption among users. Moreover, integrating predictive models with decision-support tools, such as scenario simulators and resource allocation optimizers, can help policymakers assess trade-offs and select interventions. By shifting focus from isolated models to full operational pipelines, researchers can bridge the gap between academic prototypes and real-world impact [67], [68], [83].

5.9. Scalable Big Data Architectures, Privacy, and Governance

The rise of big data in agriculture ranging from satellite imagery to mobile phone usage patterns creates opportunities for comprehensive food security monitoring but also raises challenges in data management, privacy, and governance. Future research should investigate scalable architectures capable of handling streaming, high-volume, and high-velocity data in real time [68], [83]. Distributed frameworks such as Hadoop and Spark, combined with real-time analytics platforms, could be adapted for agricultural applications. At the same time, ensuring data privacy and security is essential, especially when sensitive socio-economic information is involved. Techniques such as federated learning, secure multi-party computation, and differential privacy should be tested for food security datasets. Another critical dimension is governance: who owns the data, who has access, and how ethical use can be guaranteed. Future studies must propose frameworks that balance data-sharing for societal benefit with the protection of individual and community rights. Addressing these issues will determine whether big-data-driven risk systems can be scaled sustainably and equitably [68], [83].

5.10. Policy Integration, Cost–Benefit, and Socio-Economic Impact Analyses

Despite advances in modeling, there remains a disconnect between technical research and policy implementation. Future studies should prioritize evaluating the socio-economic impacts of deploying predictive systems, including how they affect farmer decision-making, resource allocation, and market stability [83], [67]. Cost–benefit analyses are critical to ensure that the technologies developed are economically viable, especially in resource-constrained environments. Research should also explore how predictive models can be integrated into policy frameworks, such as agricultural subsidy programs, disaster preparedness strategies, or food distribution networks. Interdisciplinary collaboration will be essential: computer scientists, agronomists, economists, and social scientists must work together to design interventions that are both technologically sound and socially relevant. Another promising direction is scenario-based policy testing, where models are used to simulate the outcomes of alternative strategies under varying conditions. Such studies will provide evidence-based guidance for governments and NGOs, ensuring that AI-driven risk assessments translate into tangible improvements in food security [83], [67].

5.11. Cross-Disciplinary Integration of Agronomic, Economic and Health Models

Food security is inherently multidisciplinary crop yields, market dynamics, nutrition outcomes, and public health are tightly coupled. Future research should prioritize the design of integrated frameworks that combine agronomic crop models, economic market simulations, and nutritional/health impact models to provide holistic risk assessments. Such integration enables tracing how production shocks propagate through markets to affect household nutrition and health outcomes. Methodologically, researchers must connect mechanistic crop/physiology models with agent-based economic models and data-driven epidemiological or nutrition models, ensuring consistent temporal and spatial resolution across components. This will require improved interoperability standards, common data schemas, and cross-domain validation exercises. Cross-disciplinary case studies coupling yield forecasts to price-response models and nutrition surveillance will generate the empirical evidence needed to guide targeted interventions, safety nets, and emergency food assistance. Work in this direction is supported by the need to move beyond siloed analyses and is aligned with studies linking ML-based nutrition forecasting and system-level thinking in food policy [69], [73], [80].

5.12. Resilience Metrics and Stress-Test Frameworks for Food Systems

Quantifying resilience, not just vulnerability, is essential for prioritizing investments and policy measures. Future work should develop standardized resilience metrics for agricultural supply chains and national food systems, and construct stress-test frameworks that evaluate system response under multiple concurrent shocks (drought + market shock + transport disruption). These frameworks should combine probabilistic scenario generation (drawing on climate projections and trade/transport vulnerability data) with system-dynamics and network analysis to identify critical nodes and failure modes. Empirical calibration of resilience indices against historical crises will improve predictive power. Decision-makers would benefit from dashboards that translate resilience scores into policy options (e.g., buffer stock release, targeted subsidies, logistics rerouting). This approach builds lessons from big-data DSS and fuzzy modeling to quantify systemic risk and prioritize adaptive investments [68], [83], [75].

5.13. Robustness and Adversarial Testing of Food-Security Models

As data-driven food security models move toward operational use, rigorous robustness testing against noisy, biased, or adversarial inputs becomes crucial. Future research should design adversarial evaluations that mimic realistic data corruption: mislabeled surveys, sensor drift, missing satellite passes, or manipulated market reports. Techniques from robust machine learning adversarial training, outlier detection, and robust statistics should be adapted to the agricultural context. Equally important are protocols for model monitoring and concept-drift detection to flag when models no longer reflect the evolving data-generating process. Benchmarks that compare robustness across model families (fuzzy hybrids, ensembles, deep nets) will help practitioners choose architectures that maintain reliable performance under stress. This line of work is particularly relevant for systems that feed into policy decisions and emergency responses [64], [68], [76].

5.14. Localized Decision Support for Smallholder Systems

Smallholder farmers make up a large share of agricultural producers in many low- and middle-income countries, but most AI systems are developed with large-scale commercial farms in mind. Future research must adapt models and decision-support tools to the realities of smallholders: high plot heterogeneity, limited data availability, low-cost devices, and specific socio-cultural constraints. This includes developing lightweight models that run offline on basic smartphones, incorporating farmer knowledge into model priors (human-in-the-loop approaches), and designing participatory evaluation protocols. Importantly, solutions should provide actionable, low-cost recommendations (timing of planting, seed choice, modest fertilizer changes) that are feasible within local resource constraints. Tailored fuzzy-DSS and region-specific benchmarks will be essential to ensure equitable benefits for smallholder systems [72], [77], [73].

5.15. Nutrient-Sensitive Food Security and Micronutrient Modeling

Food security must be broadened to include nutrient security: ensuring adequate micronutrients (iron, vitamin A, zinc, etc.) in diets. Future research should build models that map agricultural production patterns to nutrient availability and dietary intake, identifying where calorie-rich production masks micronutrient deficits. Combining production-side forecasts with household consumption surveys and nutrition surveillance enables early-warning systems that flag risks of "hidden hunger." Machine learning can predict demographic groups at highest nutritional risk under production or price shocks and help prioritize fortification, targeted transfers, or distribution of nutrient-rich crops. This research integrates nutrition-focused ML studies with broader food security modeling and supports evidence-based public-health policy [69], [80].

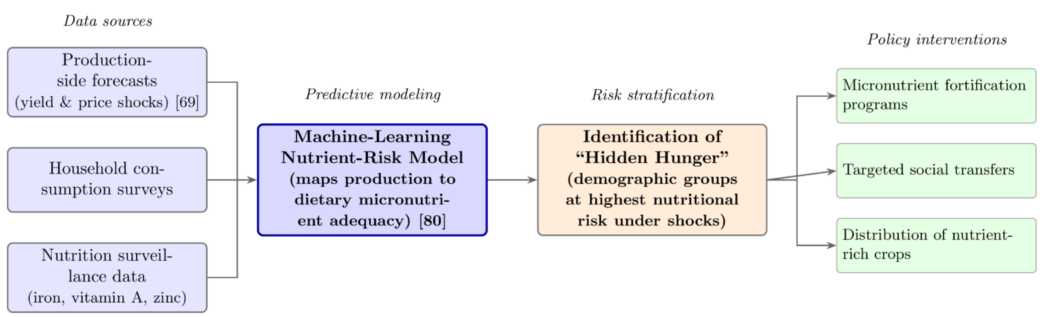

Figure 11: Nutrient-Sensitive Food Security and Micronutrient Risk Modeling Framework

5.16. Interoperable Open-Data Platforms and FAIR Data Practices

Progress depends on accessible, high-quality data. Future research must push for interoperable, open-data platforms that implement FAIR (Findable, Accessible, Interoperable, Reusable) principles for agricultural, climate, and socio-economic datasets. Work should focus on harmonizing formats (time stamps, spatial resolution, variable naming), metadata standards, and privacy-preserving access controls.

Platforms that enable federated queries (so data can be analyzed across silos without centralized pooling) will accelerate cross-regional research while respecting local governance. The development of community-maintained benchmark datasets and contest-style shared tasks will encourage reproducible science and fair comparisons across methods [64], [78], [81].

5.17. Policy-Aware Optimization and Decision-Making Under Constraints

Predictive models must be embedded in optimization frameworks that account for political, logistical, and budget constraints faced by governments. Future research should co-design algorithms that recommend interventions (e.g., targeted cash transfers, tariff adjustments, strategic reserves deployment) optimized for multiple criteria: cost-effectiveness, speed, equity, and political feasibility. Mixed-integer programming, stochastic optimization, and reinforcement learning could be adapted to propose policy bundles under uncertain scenarios; however, human oversight and interpretability are required for acceptability. Pilot studies that implement optimization outputs in controlled policy trials will help refine algorithms and demonstrate real-world impact [83], [67].

5.18. Temporal Scaling: Short-Term Alerts and Long-Term Strategic Forecasting

Food security decisions require both immediate operational alerts and long-horizon planning. Future research should develop modeling ecosystems that bridge temporal scales: short-term (weeks/months) forecasting for market and logistics decisions, seasonal forecasts for cropping choices, and decadal projections for infrastructure and adaptation planning. This will involve coupling different model classes (nowcasting with high-frequency data, seasonal statistical models, and scenario-based climate-economy projections) and designing seamless interfaces for users to navigate time horizons.

Integrating downscaled climate projections with yield and economic models enables policy planners to evaluate long-term investments in resilience [81], [82].

5.19. Ethical, Legal and Social Implications (ELSI) of AI in Food Security

Deploying AI in food systems raises ELSI concerns: data ownership, consent, algorithmic bias, and potential harm (e.g., misdirected subsidies). Future work should systematically assess these implications, develop standards for community consent and benefit sharing, and propose governance mechanisms (audit trails, impact assessments, stakeholder representation) that ensure responsible deployment. Social-science research can illuminate power dynamics in data collection and algorithmic decision-making, while legal scholars can help craft regulations that protect vulnerable actors. Linking ELSI frameworks to technical design (privacy-by-design, fairness constraints) will be essential to trustworthy systems [78], [80].

5.20. Capacity Building, End-User Training, and Adoption Pathways

Finally, technical advances will not translate into impact without capacity building. Future research should evaluate pathways for scaling solutions through training programs for extension agents, policymakers, and farmer cooperatives. Studies should identify barriers to adoption (technical literacy, trust, cost) and test different diffusion strategies: demonstration farms, mobile training modules, or public–private partnerships. Additionally, research into sustainable maintenance models who hosts and fund DSS platforms long-term is critical for persistence. Innovation technical innovation with user-centered deployment and capacity strengthening will determine whether AI-driven tools materially improve food security outcomes [68], [72], [77].

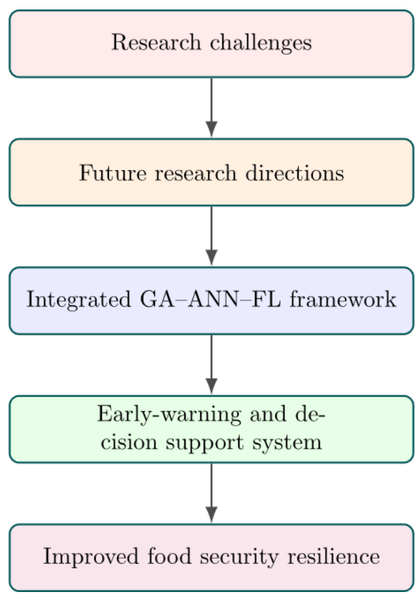

Figure 12: Conceptual Framework Linking Challenges, Research Directions and Intelligent Food Security Systems

The different strands of evidence discussed throughout this review can be synthesized into a unified conceptual perspective that links current limitations with future research opportunities.

6. Conclusion

Food security risk assessment has emerged as a critical area of research in the face of growing global challenges, including climate change, population growth, and market volatility. This review synthesized insights from a broad body of literature to evaluate the transition from traditional, static risk assessments to intelligent, dynamic systems. Specifically, it highlighted the unique synergy between Genetic Algorithms (GA), Artificial Neural Networks (ANN), and Fuzzy Logic (FL). While ANNs provide powerful predictive capabilities and GAs optimize their parameters, Fuzzy Logic bridges the gap between complex computational outputs and human reasoning by handling uncertainty and ambiguous socio-economic inputs.

Based on this synthesis, this paper proposed a conceptual roadmap for an integrated GA-ANN-Fuzzy architecture enhanced by big data protocols. However, the realization of such intelligent systems is currently hindered by several persistent challenges. Our analysis categorized these barriers into data limitations (such as temporal gaps and lack of standardization), modeling trade-offs (specifically between accuracy and interpretability), and deployment obstacles (including hardware constraints and weak adoption in low-income, highly vulnerable regions). Furthermore, the increasing frequency of climate extremes renders models trained solely on historical data unreliable.

To overcome these barriers, future research must shift from isolated algorithmic development to interdisciplinary, end-to-end decision support systems. Developing explainable AI (XAI) frameworks, lightweight edge-computing models for smallholder farmers, and climate-aware forecasting pipelines will be essential. More importantly, these technical solutions must be explicitly integrated into global and national policy frameworks. By providing probabilistic, interpretable risk outputs, the proposed integrated architecture can empower policymakers, NGOs, and agricultural stakeholders to implement proactive interventions—such as targeted subsidies, dynamic supply chain rerouting, or micronutrient fortification programs.

In conclusion, while individual AI and fuzzy computing techniques hold considerable promise, their true transformative impact lies in their integration. By advancing toward robust, interpretable, and policy-ready intelligent systems, the research community can transition food security management from reactive crisis response to proactive resilience, ultimately safeguarding vulnerable populations worldwide.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability Statement

No new data were generated or analyzed in this study. This is a literature review article and does not involve original datasets.

CRediT authorship contribution statement

Sha'elah Mohd Safri: Conceptualization, Investigation, Writing – original draft. **Muhd Khairulzaman Abd Kadir:** Supervision, Funding acquisition. **Giovanni Honakoko:** Conceptualization, Methodology, Writing – review & editing.

Declaration of Generative AI and AI-assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author(s) used Claude (Anthropic) in order to convert the manuscript from Word format into the LaTeX template required by the journal, including reformatting the reference list and reorganizing content into the required section structure. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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